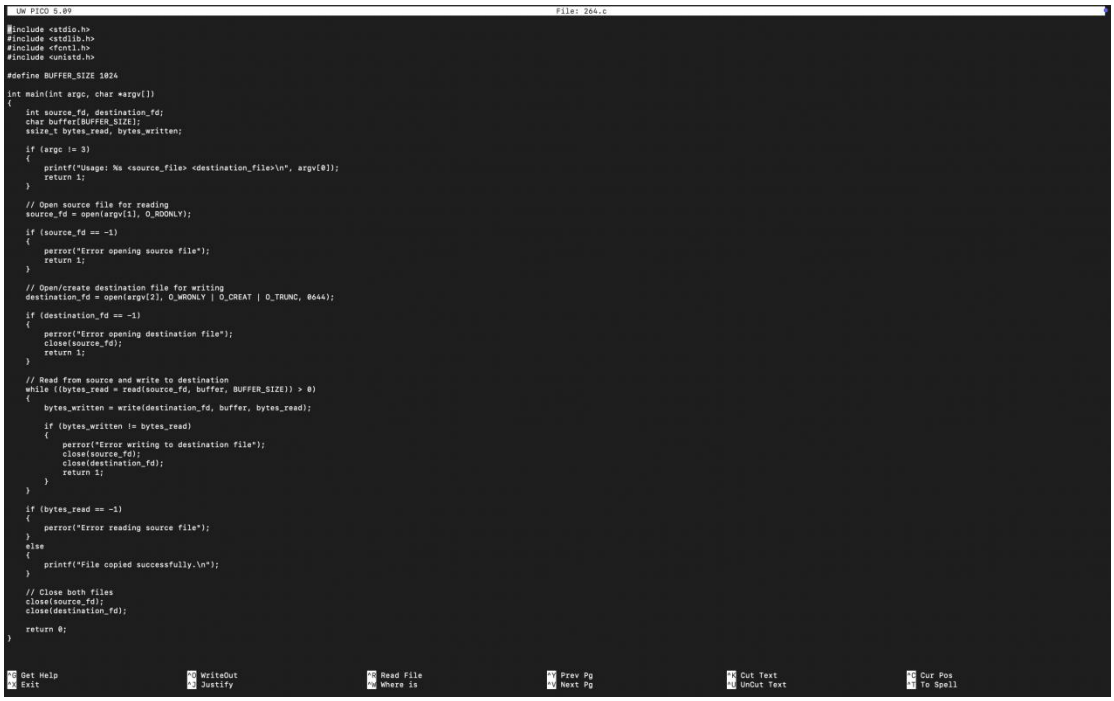
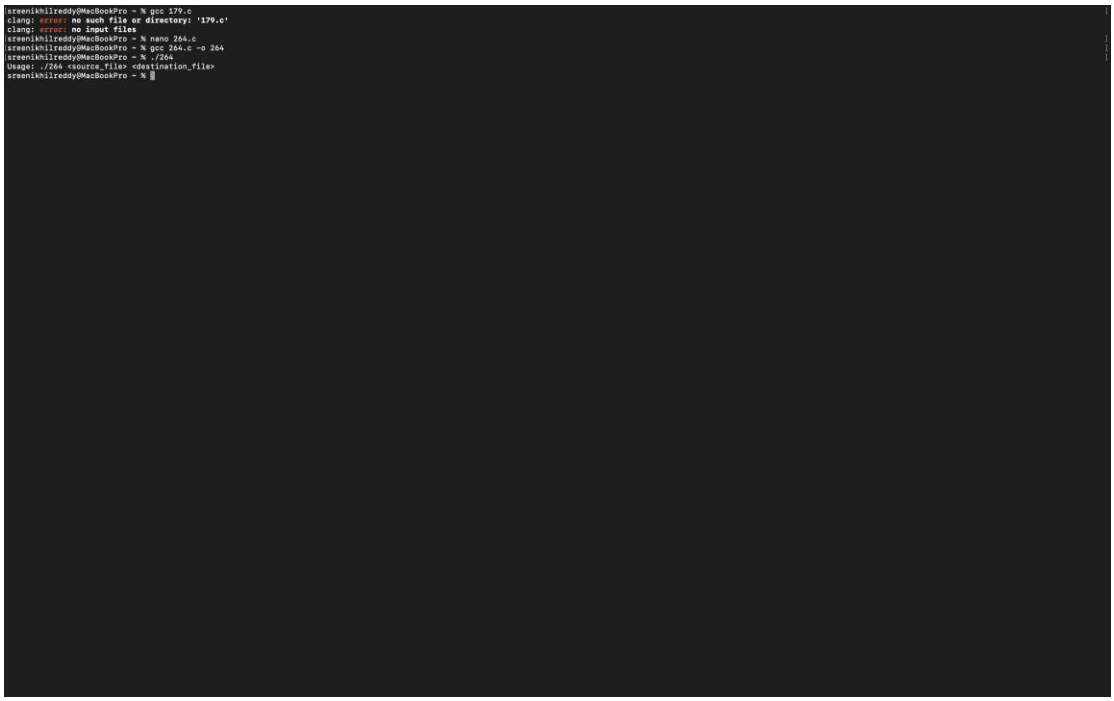
PRACTICAL WEEK 2

Task 1: Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

CODE:

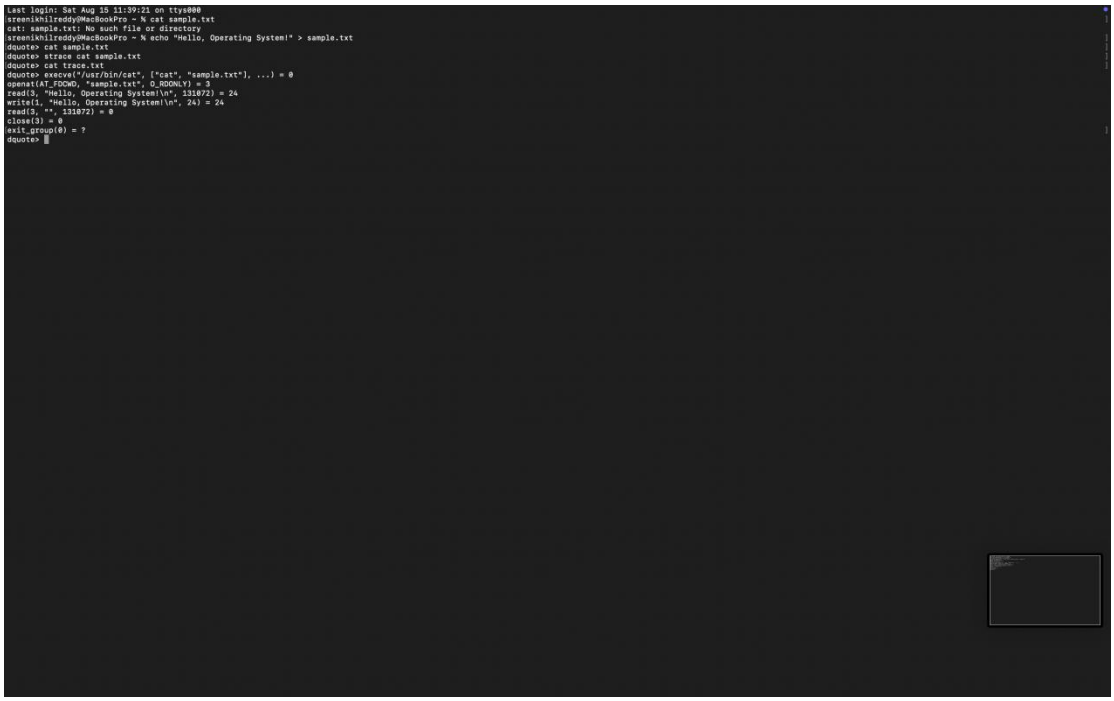


```
  UW PICO 5.09                                File: 264.c

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source_fd, destination_fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    // Open source file for reading
    source_fd = open(argv[1], O_RDONLY);

    if (source_fd == -1)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open/create destination file for writing
    destination_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination_fd == -1)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(destination_fd, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            close(source_fd);
            close(destination_fd);
            return 1;
        }
    }

    if (bytes_read == -1)
    {
        perror("Error reading source file");
    }
    else
    {
        printf("File copied successfully.\n");
    }

    // Close both files
    close(source_fd);
    close(destination_fd);

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```



```
sreenikhilreddy@MacBookPro ~ % gcc 179.c
clang: error: no such file or directory: '179.c'
clang: error: no input files
sreenikhilreddy@MacBookPro ~ % nano 264.c
sreenikhilreddy@MacBookPro ~ % gcc 264.c -o 264
sreenikhilreddy@MacBookPro ~ % ./264
Usage: ./264 <source_file> <destination_file>
sreenikhilreddy@MacBookPro ~ %
```

OUTPUT:**Task-2:** Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.



System Call	Purpose	Kernel Service
execve()	Starts the cat program	Process/program execution
openat()	Opens sample.txt	File-system access
read()	Reads file contents	File I/O
write()	Displays contents on terminal	Output/I/O
close()	Closes the file descriptor	Resource management
exit_group()	Terminates the process	Process termination